Software Requirements Specification

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Introduction

The purpose of this project is to provide a simple management system for a small restaurant business. Our goal is to make our software easier for a restaurant owner to stay organized and keep track of important business information, as many current solutions are too complex and difficult to use.

Requirements

Functional Requirements

User Interface and Navigation

- FR01 [1] The system shall provide a web-based user interface accessible through a standard browser
- FR02 [1] The system shall provide a dashboard that summarizes important restaurant information about revenue, expenses, and inventory
- FR03 [1] The system shall provide navigation for financial, inventory, and reporting functions

Expense and Revenue Tracking

- FR04 [1] The system shall allow users to create, view, edit, and delete expense records
- FR05 [1] Each expense record shall contain an amount, date, category, and description
- FR06 [1] The system shall allow users to create, view, edit, and delete revenue records
- FR07 [1] Each revenue record shall contain an amount, date, and description
- FR08 [1] The system shall calculate total revenue, total expenses, and net income for a selected period
- FR09 [2] The system shall allow users to filter financial records by date and category

Inventory Management

- FR10 [1] The system shall allow users to create, view, edit, and delete inventory items

- FR11 [1] Each inventory item shall contain a name, quantity, unit of measurement, and category
- FR12 [1] The system shall allow users to record changes to inventory quantities
- FR13 [1[The system shall maintain the current quantity of each inventory item
- FR14 [1] The system shall allow users to define a minimum quantity for inventory items and identify items that fall below that quantity
- FR15 [2] The system shall provide inventory statistics and historical information

Reporting and Analytics

- FR16 [1] The system shall generate reports summarizing revenue, expenses, and inventory information
- FR17 [2] The system shall allow users to select a time period when generating applicable reports
- FR18 [1] The system shall display calculated financial and inventory statistics in an understandable format
- FR19 [3] The system should provide charts or other visualizations for selected financial and inventory statistics

Data Management and Security

- FR20 [1] The system shall store financial and inventory information in an SQL database
- FR21 [1] The system shall validate user input before storing information in the database
- FR22 [1] The system shall provide user authentication to restrict access to business information
- FR23 [2] The system should allow users to configure categories and basic business information

Future Extensibility

- FR24 [3] The system should be structured so that additional functionality can be added without substantially modifying the core functionality

Non-functional Requirements

Usability

- NFR01 [1] The system shall be understandable and usable by users with basic computer knowledge
- NFR02 [1] The system shall use consistent navigation, terminology, and UI design
- NFR03 [1] The system shall provide clear feedback when a function works or fails
- NFR04 [1] The system shall provide meaningful error messages when users input invalid information

Performance and Reliability

- NFR05 [1] The system shall respond to normal user operations without noticeable delays under normal workload
- NFR06 [1] The system shall handle invalid input and database errors without crashing
- NFR07 [1] The system shall preserve stored information between sessions
- NFR08 [2] The system should provide a mechanism for backing up stored data

Security

- NFR09 [1] The system shall protect user information and business data from unauthorized access
- NFR10 [1] The system shall validate user input before processing or storing it
- NFR11 [1] User passwords shall not be stored in plain text

Maintainability

- NFR12 [1] The system shall use a modular code structure to separate major application components

- NFR13 [1] The source code shall follow consistent coding conventions and include documentation
- NFR14 [1] The database structure shall be documented

Compatibility and Extensibility

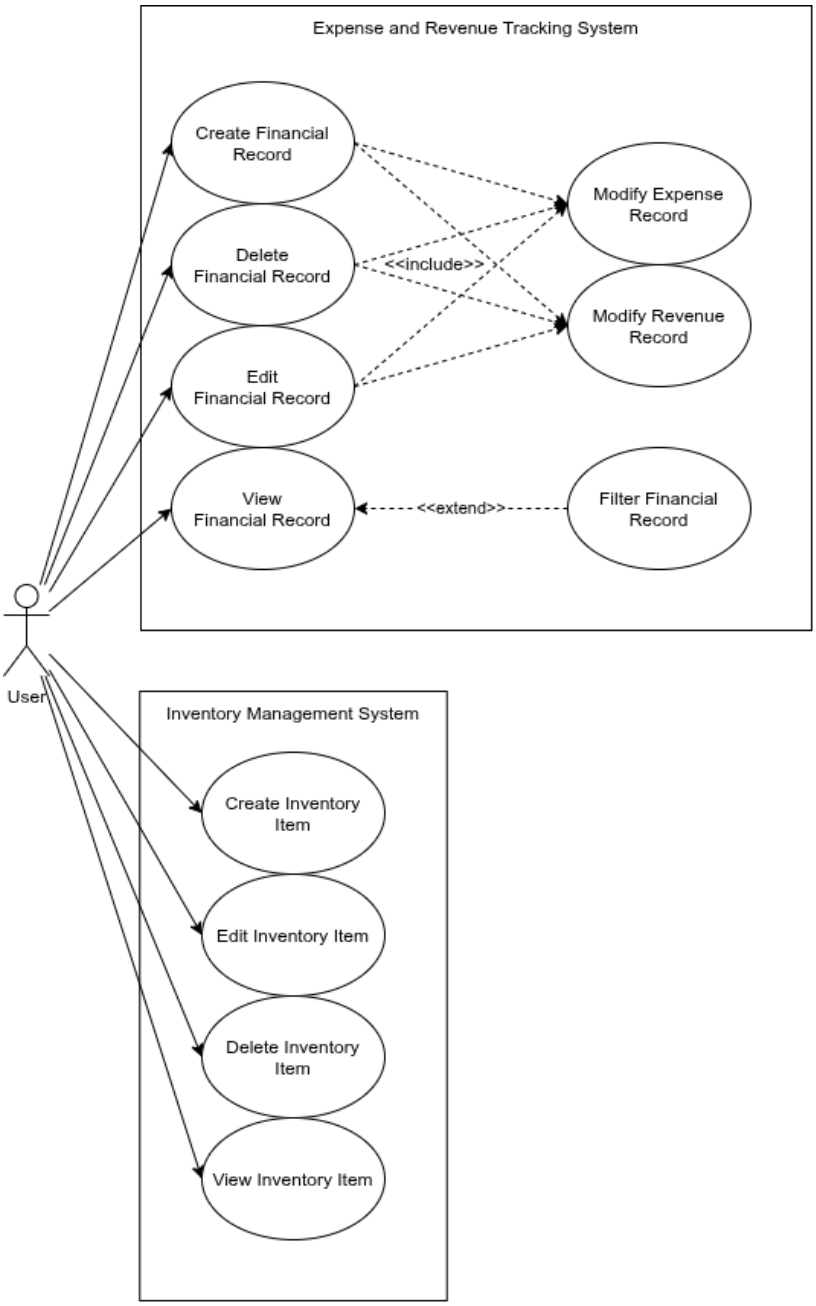
- NFR15 [1] The system shall operate through commonly used web browsers and computers
- NFR16 [1] The system shall be developed under an open-source license and include documentation describing installation and use

Use Case Modeling

Use Case Diagram

Figure 1

Illustration of Use Case Model



Detailed Use Cases

Expense and Revenue Tracking System

- UC01: Create Financial Record
    - Actor: User
    - Description: Through the web interface, the user shall be able to create a financial record, providing date, notes, costs, and inventory changes.
- UC02: Delete Financial Record
    - Actor: User
    - Description: Through the web interface, the user shall be able to delete a financial record through selecting the record and providing notes on the redaction.
- UC03: Edit Financial Record
    - Actor: User
    - Description: Through the web interface, the user shall be able to edit an existing financial record through changing one or more of the following: date, notes, costs, and inventory changes.
- UC04: Modify Expense Record
    - Actor: User
    - Description: Through the web interface, the user shall be able to interact with expense records. These need to integrate with the inventory management system and keep both the financials and the inventory totals up to date.
- UC05: Modify Revenue Record
    - Actor: User

    - Description: Through the web interface, the user shall be able to interact with revenue records. These need to integrate with the report generation system when reports are requested.
- UC06: View Financial Record
    - Actor: User
    - Description: Through the web interface, the user shall be able to view financial records saved on the system.
- UC07: Filter Financial Record
    - Actor: User
    - Description: Through the web interface, the user shall be able to filter the view of the financial records based on date range, cost range, affected inventory, or a combination of the three.

Inventory Management System

- UC08: Create Inventory Item
    - Actor: User
    - Description: Through the web interface, the user shall be able to create an inventory item, which includes the name, description, type, location, quantity, and unit price.
- UC09: Edit Inventory Item
    - Actor: User
    - Description: Through the web interface, the user shall be able to edit fields on an inventory item that exists in the system. These fields include name, description, type, location, quantity, and unit price.
- UC10: Delete Inventory Item

    - Actor: User
    - Description: Through the web interface, the user shall be able to delete an inventory item that exists in the system. This shall only be permitted to occur if the inventory is set to zero.
- UC11: View Inventory Item
    - Actor: User
    - Description: Through the web interface, the user shall be able to view an inventory item that exists within the system. This should shall all available fields for the item.

Detailed Templates

UC01: Create Financial Record

- Actor: User
- Pre-conditions: User has access to the web interface; referenced inventory items exist in the system.
- Main Flow: The user will navigate to the financial records section and select the option to create a new record. The user will then enter the date, notes, financial amounts, and applicable inventory changes. The user will then submit the form. The system will validate the information submitted and save the record to the database. The system will also update inventory totals as needed.
- Post-conditions: The financial record will be saved alongside any inventory changes. If the information is invalid, the system will display an error message and not save the record.

UC07: Filter Financial Record

- Actor: User

- Pre-conditions: User has access to the web interface; financial records can be retrieved from the system.
- Main Flow: User navigates to the financial records section and selects the filter option. The user then defines the desired date range, cost range, and affected inventory item, and then submits the criteria. The system then displays records that only match the criteria in the selected filters.
- Post-conditions: The user can view the financial records that match the selected criteria. If no records are found matching the filters, then the system displays an adequate message.

UC08: Create Inventory Item

- Actor: User
- Pre-conditions: User has access to the web interface
- Main Flow: User navigates to the inventory management section and selects the option to create a new inventory item. User then enters the item's name, description, type, location, quantity, and unit price. Then, the user submits the form. The system will then validate the information and save the new item to the database.
- Post-conditions: The inventory item will be saved to the system and usable for viewing and integrating into other systems. If information inputted is invalid, the system displays an error message and does not create the item.

Requirement Traceability Matrix

Table 1

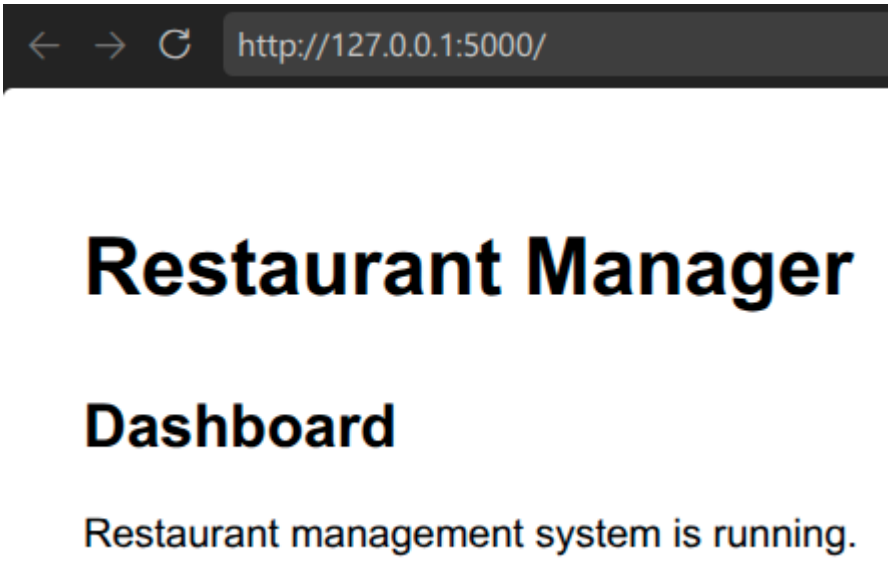
Requirement Traceability Matrix

Use Cases		UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8	UC9	UC10	UC11
Requirements	**Totals**	10	6	9	9	9	7	11	9	10	5	8
FR01	11	X	X	X	X	X	X	X	X	X	X	X
FR02	2						X					X
FR03	11	X	X	X	X	X	X	X	X	X	X	X
FR04	7	X	X	X	X	X	X	X				
FR05	5	X		X	X	X		X				
FR06	7	X	X	X	X	X	X	X				
FR07	5	X		X	X	X		X				
FR08	1							X				
FR09	1							X				
FR10	4								X	X	X	X
FR11	3								X	X		X
FR12	1									X		
FR13	1									X		
FR14	2								X	X		
FR15	1											X
FR16	0											
FR17	0											
FR18	0											
FR19	0											
FR20	11	X	X	X	X	X	X	X	X	X	X	X
FR21	7	X		X	X	X		X	X	X		

FR22	11	X	X	X	X	X	X	X	X	X	X	X
FR23	2	X							X			
FR24	0											

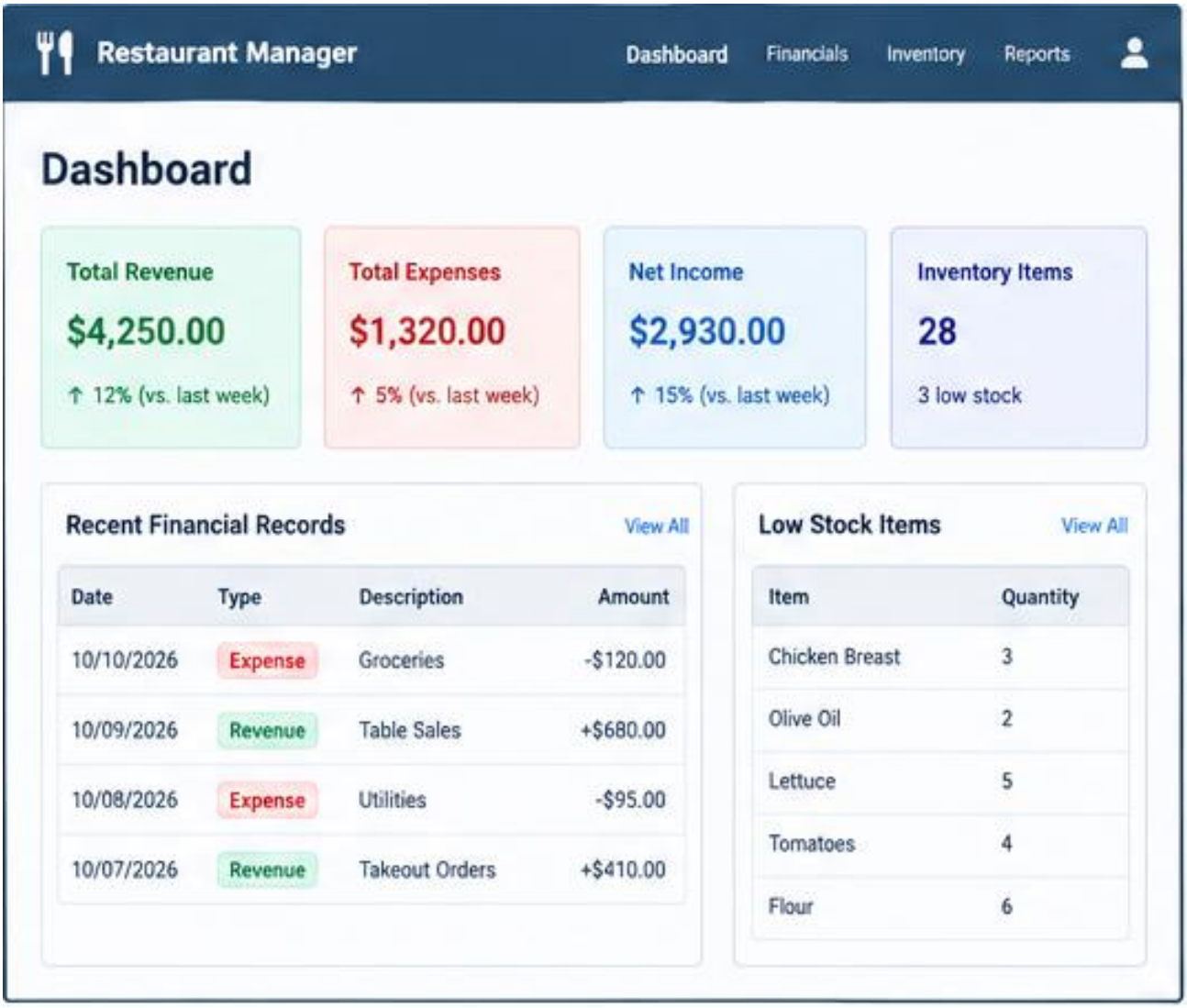
Initial Snapshots

Figure 2



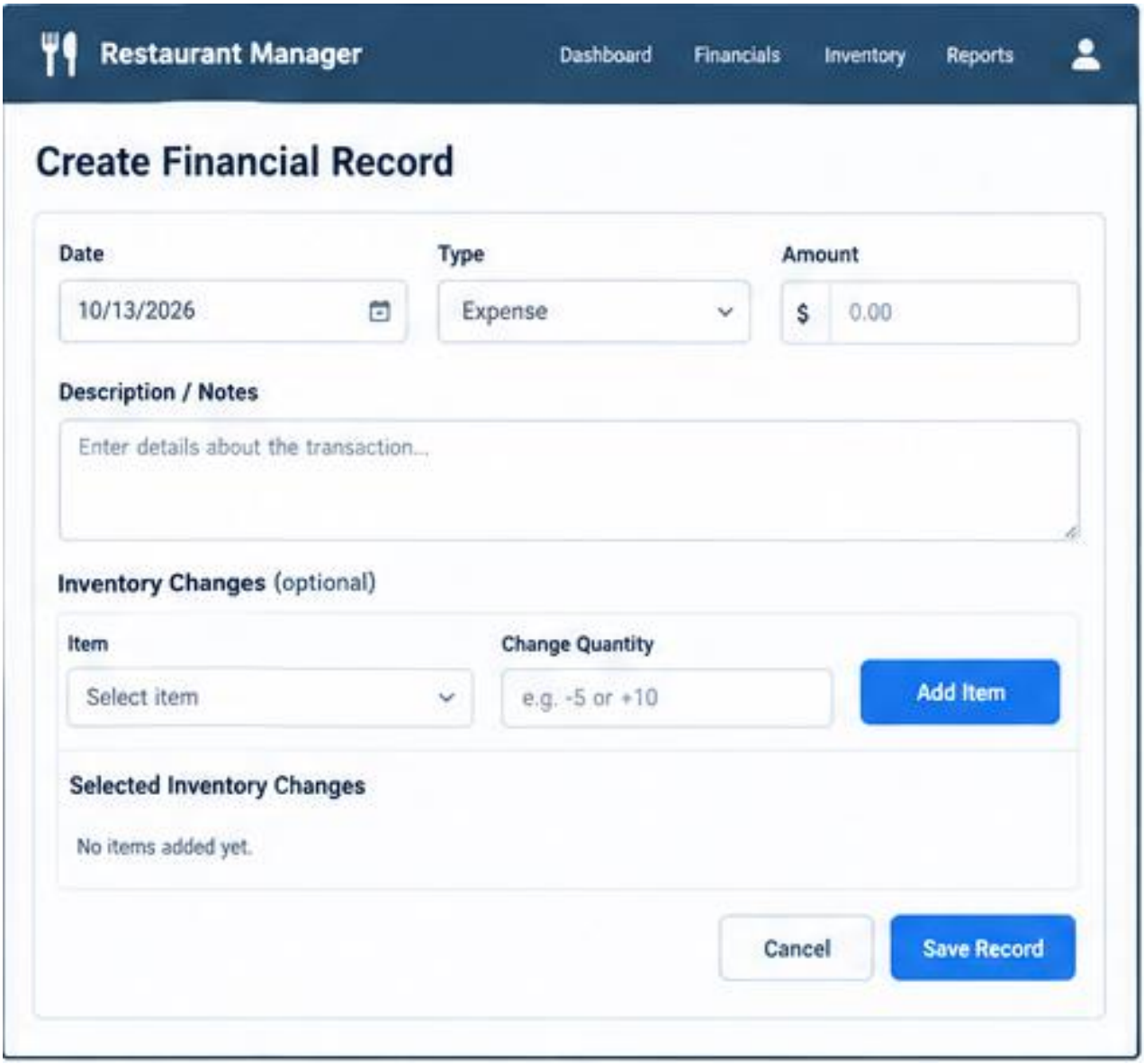
This shows our first page made for the system. This is running in Flask on a local server. We will modify CSS style sheets to make the general appearance better for the end user. Using tools to create renders, the below screenshots show a theorized version of our website.

Figure 3



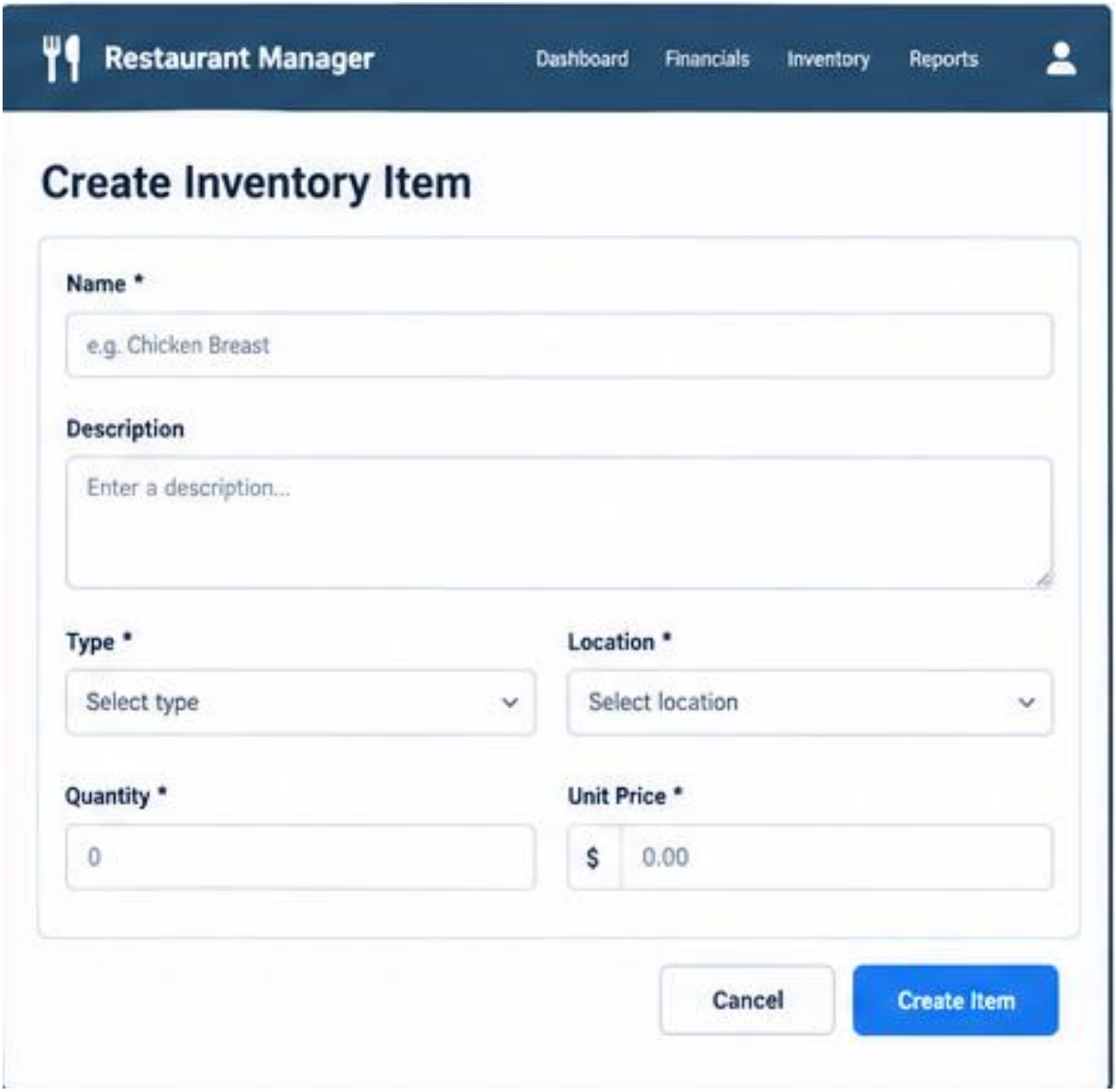
This shows a sketch of what the homepage for our website could look like. It provides the user with a quick glance at the current condition of the restaurant. Then, by clicking on view all, the user is able to view more details over anything that catches their eye.

Figure 4



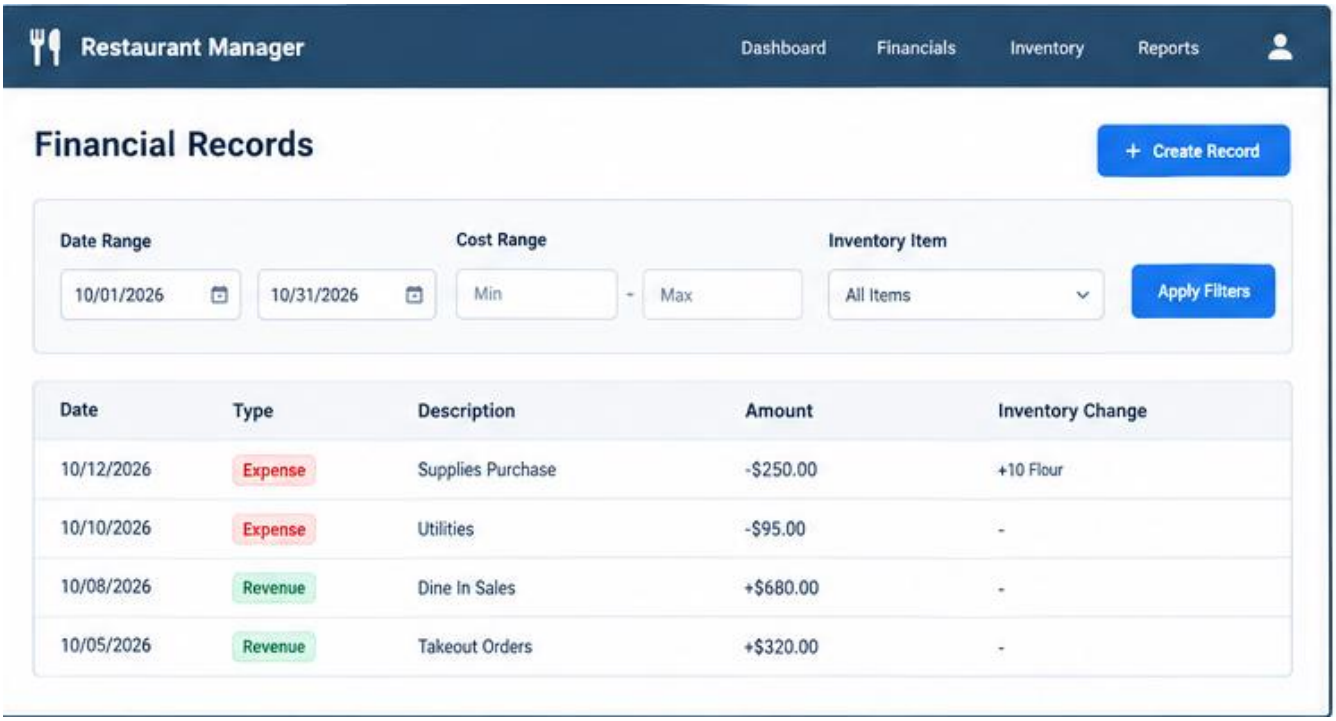
This is a template for the page the user will use to create a financial record. We've included several fields that the user can use to input details on the transaction. The user can also tie this financial record to inventory change. This ties together the inventory management system with the financial management system.

Figure 5



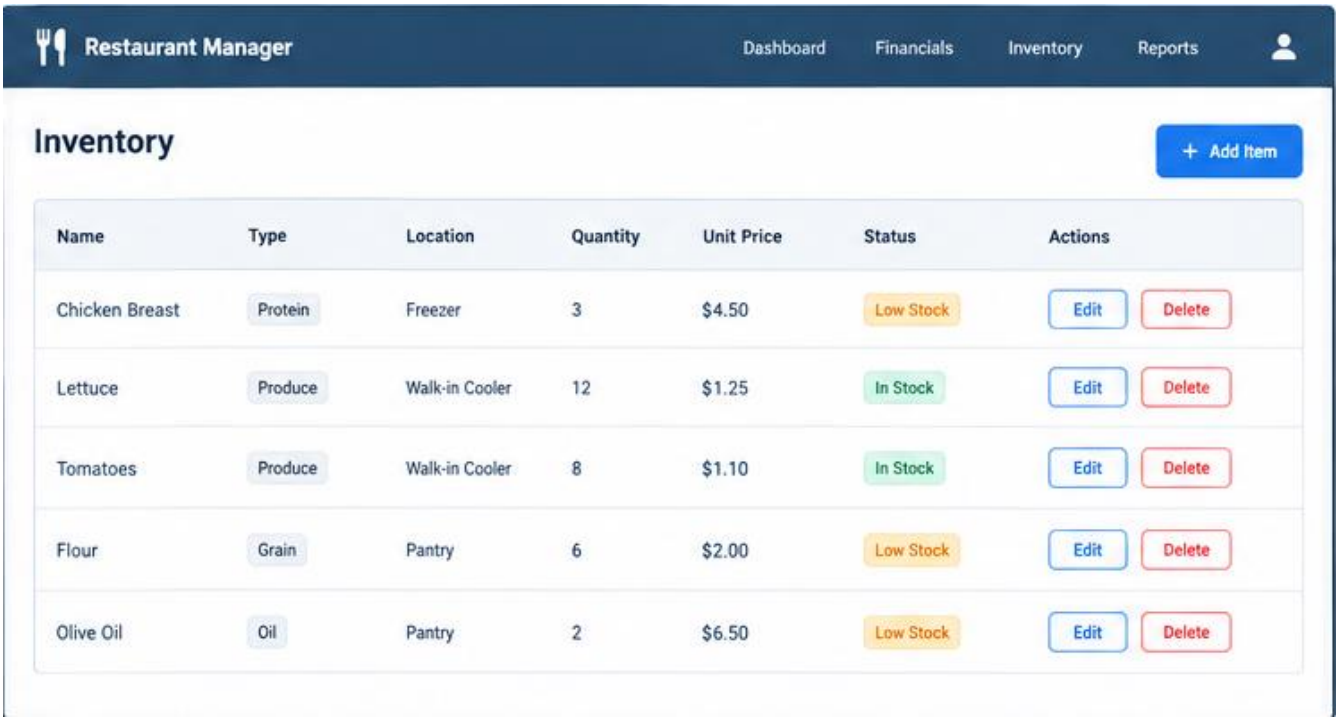
This is a sketch for the screen that allows the user to create an inventory item. The user can input data to track various inventory items, from food to cleaning supplies to office supplies. Later, we may add more fields to give more control to the user, such as what unit type to use, when to alert the user of low-stock, and supplier information.

Figure 6



Here is a sketch of the financial management system's home page. Here, you can view records and filter them as necessary. Some features we may add later are a print-friendly view, and better integration with the inventory management system.

Figure 7



Here is a snapshot of the inventory management system's home page. This is a very simple render that would match our minimum requirements. We still need to add filtering, and we may add a more graphical interface that shows the freezer, cooler, and pantry as seperate tables with their own inventory.

Glossary

Terms

- MariaDB: An open-source RDBMS commonly run in Docker.
- Docker: Collection of software that uses operating system-level virtualization to deliver software in containers.
- Container: Packages software to be run in isolated spaces in any cloud or non-cloud environment independent of type or vendor.
- Operating System-Level Virtualization: Multiple isolated user space instances being managed by the kernel of an OS.
- Kernel: A computer program that manages hardware resources being provided to applications.
- Structured Query Language (SQL): A language designed to manage data in an RDBMS.
- Relational Database Management System (RDBMS): A software system that allows users to create, maintain, and control access to a relational database.
- SQLAlchemy: An open-source Python library providing an SQL toolkit and an object-relational mapper for database interactions.
- Object-relational Mapping: Technique for converting data between a relational database and the memory of an object-oriented programming language (such as Python).
- Object-oriented programming: Code containerized based on real-world objects.
- User Interface (UI): Utilizing graphics instead of text to allow humans to interact with machines.
- Flask: A micro web framework that is written in Python that is extendable through plugins.
- Python: A high-level programming language built for ease-of-implementation with an extensive standard library.

- Cascading Style Sheets (CSS): A language for specifying the presentation of a document written in HTML.
- Hypertext Markup Language (HTML): The standard markup language for documents designed for viewing in a web browser.

References

- https://www.netsuite.com/portal/resource/articles/inventory-management/restaurant-inventory-management.shtml
- https://www.upmenu.com/blog/restaurant-management-software/

Similar Projects

- https://github.com/harismuneer/Restaurant-Management-System

Contributions

Cole Raymond:

- Introduction
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- Glossary

Jorden Fisher:

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Julius Murillo: